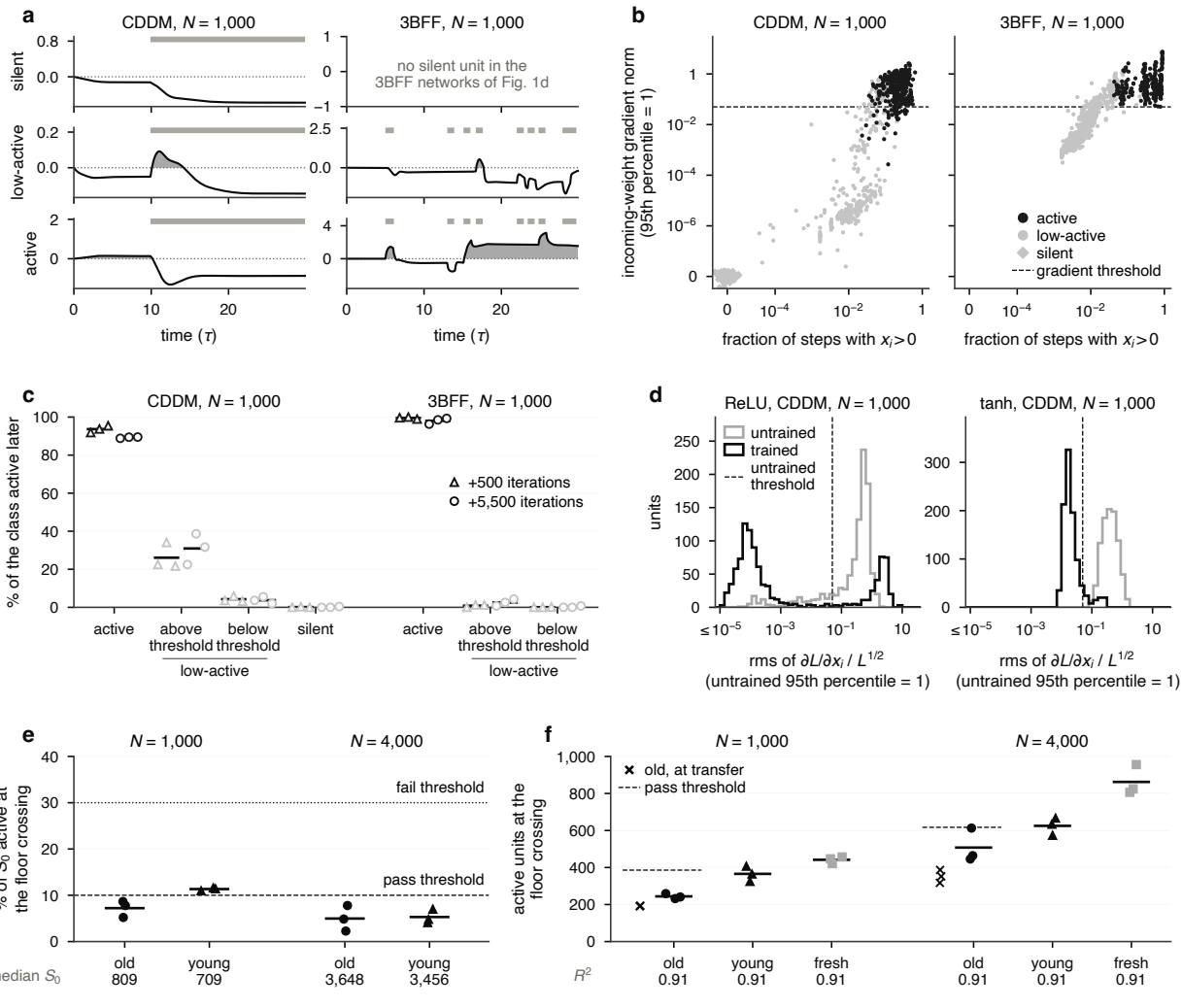

a
CDDM, N = 1,000
3BFF, N = 1,000
silent
low-active
active
no silent unit in the
3BFF networks of Fig. 1d
time (τ)
b
CDDM, N = 1,000
3BFF, N = 1,000
incoming-weight gradient norm
(95th percentile = 1)
active
low-active
silent
gradient threshold
fraction of steps with $x_i > 0$
c
CDDM, N = 1,000
3BFF, N = 1,000
% of the class active later
+500 iterations
+5,500 iterations
active
above threshold
below threshold
silent
low-active
d
ReLU, CDDM, N = 1,000
tanh, CDDM, N = 1,000
units
untrained
trained
untrained threshold
rms of $\partial L/\partial x_i$ / $L^{1/2}$
(untrained 95th percentile = 1)
e
N = 1,000
N = 4,000
% of S_0 active at the floor crossing
fail threshold
pass threshold
median S_0
old 809
young 709
old 3,648
young 3,456
f
N = 1,000
N = 4,000
active units at the floor crossing
old, at transfer
pass threshold
R^2
old 0.91
young 0.91
fresh 0.91
old 0.91
young 0.91
fresh 0.91